Mr. Haodong Chen

Email: haodong.chen1@student.uq.edu.au
[Homepage] [GitHub] [Google Scholar]

Profile

I am a second-year PhD candidate at the Information Engineering Lab (ieLab), The University of Queensland, supervised by Dr. Teerapong Leelanupab and Prof. Guido Zuccon. My research focuses on **optimising retrieval for deep research agents**: both building retrieval systems that better serve agents and helping agents make better use of retrieval. I am also interested in LLM-based ranking and retrieval-augmented generation.

Education background

The University of Queensland (UQ) **Brisbane, Australia**
- *Doctor of Philosophy in Computer Science* **07/2025-Now**
- Research Interests: Information Retrieval, Long-Horizon Search, Deep Research Agents
- Supervisor: Dr. Teerapong Leelanupab and Prof. Guido Zuccon

The University of Queensland (UQ) **Brisbane, Australia**
- *Master of Engineering Science in Software Engineering* **07/2024-07/2025**
- Research Interests: Retrieval-Augmented Generation, LLM Agents, Biomedical Question Answering
- GPA: 6.75/7.0

Guangzhou City University of Technology **Guangzhou, China**
- *Bachelor of Engineering in Software Engineering* **09/2019-06/2023**
- GPA: 3.41/4.0

Publications

1. **Chen H.**, Wang S., Yin Y., Zhuang S., Zuccon G. & Leelanupab T. (2027). ITER: Interaction-Aware Retrieval for Agentic Search. *WSDM 2027 Findings*. [Paper]

2. **Chen H.**, Zhuang S., Yao Z., Zuccon G. & Leelanupab T. (2026). Where Relevance Emerges: A Layer-Wise Study of Internal Attention for Zero-Shot Re-Ranking. *SIGIR 2026*. [Paper]

3. **Chen H.**, Zuccon G. & Leelanupab T. (2025). Beyond GeneGPT: A Multi-Agent Architecture with Open-Source LLMs for Enhanced Genomic Question Answering. *SIGIR-AP 2025*. [Paper]

4. Wang S., **Chen H.**, Yin Y., Zhuang S., Koopman B. & Zuccon G. (2026). Search, Inspect, Fetch: Exploiting Structure-Aware Boolean Retrieval for Deep-Search Agents. *arXiv:2608.02751*. (Under Review) [Paper]

Work Experience

The University of Queensland (UQ) **Brisbane, Australia**
Tutor, School of Electrical Engineering and Computer Science **02/2025-Now**
- DATA7901 / DATA7903 Data Science Capstone Project
- INFS7203 Data Mining

Fits (Guangdong) Technology Co., Ltd. **Guangzhou, China**
Software Engineer **10/2022-06/2024**

Awards

CCIR Cup 2026 (China Conference on Information Retrieval): Second Prize **08/2026**
Track: Retrieval Model Optimization Based on Agent Interaction Trajectories
Dean's Commendation for Academic Excellence: The University of Queensland **07/2025**

Technical Skills

- **Programming skills:** Python (PyTorch, Transformers, vLLM), Java, C, JavaScript/TypeScript, SQL and AI coding tools.
- **Language skills:** Mandarin (native), Cantonese (native) and English (fluent).